# Fengbei Liu

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## Education

**Cornell Tech, Cornell University** — *New York City, NY, US*
*Postdoctoral Researcher;* — *Feb 2024 – Present*
*Supervisor: Prof. Mert R. Sabuncu*

**Australian Institute for Machine Learning (AIML)** — *Adelaide, Australia*
*Ph.D in Computer Vision and Medical Imaging;* — *Apr 2020 – Oct 2023*
*Supervisor: Prof. Gustavo Carneiro, Prof. Mark Jenkinson*

**University of Adelaide** — Adelaide, Australia
*Honors in Computer Science; (First Class)* — *Mar 2019 – Mar 2020*

**University of Adelaide** — Adelaide, Australia
*Bachelor in Computer Science;* — *Jul 2015 – Jul 2018*

## Research Interests

**Keywords: Computer Vision, Medical Image Analysis, Weakly-supervised Learning**

My research interest is in computer vision and medical image anlaysis. I focus on developing weakly-supervised deep learning solutions and its application in medical imaging, including noisy label learning, semi-supervised learning, partial-label learning and active learning.

## Publications

### First-author Publications

**Bridging Generative and Discriminative Noisy-Label Learning via Direction-Agnostic EM Formulation**
***Fengbei Liu***, *Chong Wang, Yuanhong Chen, Yuyuan Liu, Gustavo Carneiro*
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2026

**MAdam: Metric-Aware Multi-Objective Adam**
***Fengbei Liu***, *Rachit Saluja, Sunwoo Kwak, Ruibo Wang, Ruining Deng, Heejong Kim, Johannes C. Paetzold, Mert R. Sabuncu*
NeurIPS 2026

**RNED: Rotary Number Encoding and Decoding for Quantitative Medical VLM Analysis**
***Fengbei Liu***, *Sunwoo Kwak, Nusrat Binta Nizam, Ilan Richter, Ashley Beecy, Jayant K Raikhelkar, Deborah Estrin, Mert R. Sabuncu*
CVPR 2026

**HyperCT: Low-Rank Hypernet for Unified Chest CT Analysis**
***Fengbei Liu***, *Sunwoo Kwak, Hao Phung, Nusrat Binta Nizam, Ilan Richter, Nir Uriel, Hadar Averbuch-Elor, Deborah Estrin, Mert R. Sabuncu*
MIDL 2026

**BoMD: Bag of Multi-label Local Descriptors for Noisy Chest X-ray Classification**
*Yuanhong Chen*, **Fengbei Liu***, Hu Wang, Chong Wang, Yuyuan Liu, Yu Tian, Gustavo Carneiro*
ICCV 2023

**Self-supervised Multi-class Pre-training for Unsupervised Anomaly Detection and Segmentation in Medical Images**
*Yu Tian*, **Fengbei Liu***, Guansong Pang, Yuanhong Chen, Yuyuan Liu, Johan W. Verjans, Rajvinder Singh, Gustavo Carneiro*
Medical Image Analysis (MedIA) 2023

**NVUM: Non-volatile Unbiased Memory for Robust Medical Image Classification**
***Fengbei Liu***, *Yuanhong Chen, Yu Tian, Yuyuan Liu, Chong Wang, Vasileios Belagiannis, Gustavo Carneiro*
MICCAI 2022, Early Accept, featured in ComputerVisionNews

**ACPL: Anti-curriculum Psueod-labelling for Semi-supervised Medical Image Classification**
***Fengbei Liu****, Yu Tian*, *Yuanhong Chen, Yuyuan Liu, Vasileios Belagiannis, Gustavo Carneiro*
CVPR 2022

**Self-supervised Mean-teacher for Semi-supervised Chest X-ray Classification**
***Fengbei Liu****, Yu Tian, Filipe R. Cordeiro, Vasileios Belagiannis, Ian Reid, Gustavo Carneiro*
MICCAI-MLMI 2021

**Self-supervised Depth estimation to Regularise Semantic Segmentation in Knee Arthroscopy**
***Fengbei Liu****, Yaqub Jonmohamadi, Gabriel Maicas, Ajay K Pandey, Gustavo Carneiro*
MICCAI 2020

## Co-author Publications

**Deep Learning With Longitudinal Electrocardiograms Predicts Atrial Fibrillation Better Than a Single Electrocardiogram**
*Sunwoo Kwak,* ***Fengbei Liu****, Nusrat B. Nizam, Peter M. Okin, Mert R. Sabuncu*
JACC: Advances 2026, Accepted October 15

**MedCure: Medical Data Curation for Efficient Vision-Language Pretraining**
*Chong Wang,* ***Fengbei Liu****, Yuyuan Liu, Yuanhong Chen, Yu Tian, Leyuan Fang*
IEEE Transactions on Image Processing (TIP) 2026

**BackSplit: The Importance of Sub-dividing the Background in Biomedical Lesion Segmentation**
*Rachit Saluja, Asli Cihangir, Ruining Deng, Johannes C. Paetzold,* ***Fengbei Liu****, Mert R. Sabuncu*
CVPR 2026

**From Failure to Feedback: Group Revision Unlocks Hard Cases in Object-Level Grounding**
*Yuyuan Liu, Yijing Ji, Anjie Le, Jiayuan Zhu, Jiazhen Pan, Can Peng, Jiajun Deng,* ***Fengbei Liu****, Junde Wu*
CVPR 2026

**Beyond Machine Interpretation: Learning from Expert Over-Reads Improves ECG Diagnosis**
*Sunwoo Kwak,* ***Fengbei Liu****, Nusrat Binta Nizam, Ilan Richter, Nir Uriel, Peter M. Okin, Mert R. Sabuncu*
MIDL 2026

**X-Cardia: Phenotype-Guided Cross-Modal Alignment for Opportunistic Cardiac Screening on Routine Chest CT**
*Nusrat Binta Nizam,* ***Fengbei Liu****, Sunwoo Kwak, Ilan Richter, Jayant K Raikhelkar, Ashley Beecy, Nir Uriel, Deborah Estrin, Mert R. Sabuncu*
MIDL 2026, Oral

**ShapKO: Shapley-Adaptive Modality Knockout for Robust Multimodal Learning**
*Nusrat Binta Nizam,* ***Fengbei Liu****, Sunwoo Kwak, Minh Nguyen, Ruining Deng, Mert R. Sabuncu*
MICCAI 2026

**An Artificial Intelligence Model to Detect Abnormal Ejection Fraction from Non-Contrast Chest Computed Tomography: The CT-LVEF study**
*Jayant Raikhelkar, Zilong Bai, Ashley N Beecy, Ilan Richter,* ***Fengbei Liu****, et al.*
European Heart Journal - Digital Health (EHJ-DH) 2026

**Cardio Amyloid-AI: Advanced Multimodal Screening for Transthyretin Cardiac Amyloidosis in Severe Aortic Stenosis Patients**
*Nusrat Binta Nizam, Ashley N Beecy, Jacob W Groenendyk,* ***Fengbei Liu****, et al.*
European Heart Journal - Digital Health (EHJ-DH) 2026

**Progressive Mining and Dynamic Distillation of Hierarchical Prototypes for Disease Classification and Localisation**
*Chong Wang,* ***Fengbei Liu****, Yuanhong Chen, Gustavo Carneiro*
IEEE Journal of Biomedical and Health Informatics (JBHI) 2025

**Cross- and Intra-image Prototypical Learning for Multi-label Disease Diagnosis and Interpretation**
*Chong Wang,* ***Fengbei Liu****, Yuanhong Chen, Helen Frazer, Gustavo Carneiro*
IEEE Transactions on Medical Imaging (TMI) 2025

**Mixture of Gaussian-distributed Prototypes with Generative Modelling for Interpretable and Trustworthy Image Recognition**
*Chong Wang, Yuanhong Chen,* ***Fengbei Liu****, Yuyuan Liu, Davis J McCarthy, Helen Frazer, Gustavo Carneiro*
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2025

**Knockout: A Simple Way to Handle Missing Inputs**
*Minh Nguyen, Batuhan K. Karaman, Heejong Kim, Alan Q. Wang,* ***Fengbei Liu****, Mert R. Sabuncu*
Transactions on Machine Learning Research (TMLR) 2025

**Translation Consistent Semi-supervised Segmentation for 3D Medical Images**
*Yuyuan Liu, Yu Tian, Chong Wang, Yuanhong Chen,* ***Fengbei Liu****, Vasileios Belagiannis, Gustavo Carneiro*
IEEE Transactions on Medical Imaging (TMI) 2024

**BRAIxDet: Learning to Detect Malignant Breast Lesion with Incomplete Annotations**
*Yuanhong Chen, Yuyuan Liu, Chong Wang, Michael Elliott, Chun Fung Kwok, Yu Tian,* ***Fengbei Liu****, Helen Frazer, Davis J McCarthy, Gustavo Carneiro*
Medical Image Analysis (MedIA) 2024

**Unraveling Instance Associations: A Closer Look for Audio-Visual Segmentation**
*Yuanhong Chen, Yuyuan Liu, Hu Wang,* ***Fengbei Liu****, Chong Wang, Helen Frazer, Gustavo Carneiro*
CVPR 2024

**Effective Segmentation of Post-Treatment Gliomas Using Simple Approaches: Artificial Sequence Generation and Ensemble Models**
*Heejong Kim, Léo Milecki, Mina C. Moghadam,* ***Fengbei Liu****, Minh Nguyen, Eric Qiu, Abhishek Thanki, Mert R. Sabuncu*
MICCAI BraTS Challenge 2024, Oral Presentation

**An Interpretable and Accurate Deep-learning Diagnosis Framework Modelled with Fully and Semi-supervised Reciprocal Learning**
*Chong Wang, Yuanhong Chen,* ***Fengbei Liu****, Michael Elliott, Chun Fung Kwok, Carlos Peña-Solorzano, Helen Frazer, Davis James McCarthy, Gustavo Carneiro*
IEEE Transactions on Medical Imaging (TMI) 2023

**Learning Support and Trivial Prototypes for Interpretable Image Classification**
*Chong Wang, Yuyuan Liu, Yuanhong Chen,* ***Fengbei Liu****, Yu Tian, Davis J McCarthy, Helen Frazer, Gustavo Carneiro*
ICCV 2023

**Unsupervised anomaly detection in medical images with a memory-augmented multi-level cross-attentional masked autoencoder**
*Yu Tian, Guansong Pang, Yuyuan Liu, Chong Wang, Yuanhong Chen,* ***Fengbei Liu****, Rajvinder Singh, Johan W Verjans, Gustavo Carneiro*
MICCAI-MLMI 2023

**Knowledge Distillation to Ensemble Global and Interpretable Prototype-Based Mammogram Classification Models**
*Chong Wang, Yuanhong Chen, Yuyuan Liu, Yu Tian,* ***Fengbei Liu*** *Davis J McCarthy, Michael Elliott, Helen Frazer, Gustavo Carneiro*
MICCAI 2022, Early Accept

**Multi-view Local Co-occurrence and Global Consistency Learning Improve Mammogram Classification Generalisation**
*Yuanhong Chen, Hu Wang, Chong Wang, Yu Tian,* ***Fengbei Liu****, Yuyuan Liu, Michael Elliott, Davis J McCarthy, Helen Frazer, Gustavo Carneiro*
MICCAI 2022, Early Accept

**Contrastive Transformer-based Multiple Instance Learning for Weakly Supervised Polyp Frame Detection**
*Yu Tian, Guansong Pang,* ***Fengbei Liu****, Yuyuan Liu, Chong Wang, Yuanhong Chen, Johan W Verjans, Gustavo Carneiro*
MICCAI 2022, Early Accept

**Perturbed and Strict Mean Teachers for Semi-supervised Semantic Segmentation**
*Yuyuan Liu, Yu Tian, Yuanhong Chen,* ***Fengbei Liu****, Belagiannis Vasileios, Gustavo Carneiro*
CVPR 2022

**Pixel-wise Energy-biased Abstention Learning for Anomaly Segmentation on Complex Urban Driving Scenes**
*Yu Tian*, Yuyuan Liu*, Guansong Pang,* ***Fengbei Liu****, Yuanhong Chen, Gustavo Carneiro*
ECCV 2022, Oral

**Constrained Contrastive Distribution Learning for Unsupervised Anomaly Detection and Localisation in Medical Images**
*Yu Tian, Guansong Pang, **Fengbei Liu**, Yuanhong Chen, Seon Ho Shin, Johan W Verjans, Rajvinder Singh, Gustavo Carneiro*
MICCAI 2021

**3D Semantic Mapping from Arthroscopy using Out-of-distribution Pose and Depth and In-distribution Segmentation Training**
*Yaqub Jonmohamadi, Shahnewaz Ali, **Fengbei Liu**, Jonathan Roberts, Ross Crawford, Gustavo Carneiro, Ajay K Pandey*
MICCAI 2021

**Automatic Segmentation of Multiple Structures in Knee Arthroscopy Using Deep Learning**
*Yaqub Jonmohamadi, Yu Takeda, **Fengbei Liu**, Fumio Sasazawa, Gabriel Maicas, Ross Crawford, Jonathan Roberts, Ajay K. Pandey, Gustavo Carneiro*
IEEE Access 2020

## Preprint/Under-Review

**Agentic Large-Language-Model Systems in Medicine: A Systematic Review and Taxonomy**
*Abdul Mohaimen Al Radi, Xu Cao, Fanyang Yu, Yuyuan Liu, **Fengbei Liu**, Chong Wang, Yuanhong Chen, Jintai Chen, Hu Wang, Yanda Meng, Zhenyi Wang, Chen Chen, Mubarak Shah, Tianyu Han, Christos Davatzikos, MacLean P. Nasrallah, Yu Tian*
TechRxiv, 2025

**Asymmetric Co-teaching with Multi-view Consensus for Noisy Label Learning**
***Fengbei Liu**, Yuanhong Chen, Chong Wang, Yu Tian, Gustavo Carneiro*
arXiv:2301.01143, 2023

## Professional Activities

### Conference

**Reviewer:** ICCV 2021/2023, MICCAI 2021/2026, CVPR 2022/2023/2024/2026, ECCV 2022/2024, BMVC 2022, NeurIPS 2023/2024/2026, ICLR 2024, ICML 2024

### Journal

**Reviewer:** IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

## Awards & Achievements

**Ph.D Scholarship:** University of Adelaide, Apr 2020

## Skills

**Programming:** Python & Pytorch, Slurm, Docker & Kubernetes

**Language:** Mandarin (native), English (professional)

## References

**Prof. Gustavo Carneiro**
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**Prof. Vasileios Belagiannis**
*Professor, Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)*
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**Prof. Mert R. Sabuncu**
*Professor, Cornell University, Cornell Tech and Weill Cornell Medicine*
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